برنامج هندسة النظم الذكية

Program of Intelligent Systems Engineering

جداول الخطة الدراسية

Study Plan

Program of Intelligent Systems Engineering (ISE)				برنامج هندسة النظم الذكية					
Level 0 – Semester 1				المستوى صفر– الفصل الدراسي الأول					
Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
MEC0001	Production Engineering	None	3	2	-	3	5		هندسة انتاج
GEN0002	Mathematics (1)	None	3	2	2	-	4		رياضيات (1)
GEN0003	Physics (1)	None	3	2	-	3	5		فيزياء (1)
GEN0004	Mechanics	None	3	2	2	-	4		ميكانيكا
GEN0005	Academic Reading & Writing (1)	None	2	2	-	-	2		القراءة والكتابة الأكاديمية (1)
GEN0006	Computer Skills	None	2	2	-	-	2		مهارات الحاسب الالي
Total			**16**	**12**	**4**	**6**	**22**		**المجموع**

Program of Intelligent Systems Engineering (ISE)				برنامج هندسة النظم الذكية					
Level 0 – Semester 2				المستوى صفر– الفصل الدراسي الثانى					
Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
GEN0007	Mathematics (2)	GEN0002	3	2	2	-	4		رياضيات (2)
GEN0008	Physics (2)	GEN0003	3	2	-	3	5		فيزياء (2)
GEN0009	Industrial Chemistry	None	3	2	-	2	4		كيمياء صناعية
MEC0010	Engineering Drawing	None	3	2	3	-	5		رسم هندسي
GEN0011	Academic Reading & Writing (2)	GEN0005	2	2	-	-	2		القراءة والكتابة الأكاديمية (2)
GEN0012	Critical Thinking	None	2	2	-	-	2		التفكير النقدي
Total			**16**	**12**	**5**	**5**	**22**		**المجموع**

Program of Intelligent Systems Engineering (ISE)				برنامج هندسة النظم الذكية					
Level 1 – Semester 1				المستوى الأول – الفصل الدراسي الأول					
Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
GEN1601	Mathematics (3)	GEN0007	3	2	2	-	4		رياضيات (3)
POW1602	Electrical Engineering	GEN0008	3	2	2	-	4		هندسة كهربية
ECE1603	Analog Electronics	None	3	2	-	3	5		الكترونيات تناظرية
CSE1604	Data Oriented Programming Language	GEN0006	3	2	-	3	5		لغات برمجة بيانات
GEN1006	Foundations of Digital Technology	None	2	2	-	-	2		أسس التكنولوجيا الرقمية
GEN1007	Professional Ethics	None	2	2	-	-	2		أخلاقيات المهنة
Total			**16**	**12**	**4**	**6**	**22**	**المجموع**	

Program of Intelligent Systems Engineering (ISE)				برنامج هندسة النظم الذكية					
Level 1 – Semester 2				المستوى الأول - الفصل الدراسي الثانى					
Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
ECE1605	Digital Electronics	ECE1603	3	2	-	3	5		الكترونيات رقمية
ECE1606	Analog Communications	None	3	2	-	2	4		اتصالات تناظرية
ECE1607	Sensors and Actuators	ECE1603	2	2	-	1	3		المستشعرات والمفعلات
ECE1608	Electromagnetic	GEN0008	3	2	2	-	4		كهرومغناطيسية
GENXXXX	Elective 1 (UR)	None	2	2	-	-	2		اختياري 1 (م ج)
------	Elective 1 (DR)	None	3	2	2	-	4		اختياري 1 (م.م)
Total			**16**	**12**	**4**	**6**	**22**		**المجموع**

Program of Intelligent Systems Engineering (ISE)				برنامج هندسة النظم الذكية					
Level 2 – Semester 1				المستوى الثانى - الفصل الدراسي الأول					
Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
ECE2601	Digital Signal Processing	ECE1606	3	2	2	-	4		معالجة إشارات رقمية
ECE2602	Probability and Statistics	GEN0007	3	2	-	2	4		الاحتمالات والاحصاء
POW2603	Theory of Electrical Machines	POW1602	2	2	-	-	2		نظرية آلات كهربية
CSE2604	Data Structures and Algorithms	CSE1604	3	2	-	2	4		هياكل البيانات والخوارزميات
CSE2605	Operating Systems	CSE1604	2	2	-	-	2		نظم التشغيل
GENXXXX	Elective 2 (UR)	None	2	2	-	-	2		اختياري 2 (م ج)
------	Elective 2 (DR)	------	3	2	2	-	4		اختياري 2 (م.م)
Total			**18**	**14**	**4**	**4**	**22**		**المجموع**

Program of Intelligent Systems Engineering (ISE)				برنامج هندسة النظم الذكية					
Level 2 – Semester 2				المستوى الثانى - الفصل الدراسي الثانى					
Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
ECE2606	Digital Communications	ECE1606	3	2	-	2	4		الاتصالات الرقمية
ECE2607	Embedded Systems	None	3	2	-	2	4		الأنظمة المدمجة
CSE2608	Linear Systems and Control	GEN1601	3	2	2	-	4		الأنظمة الخطية والتحكم
CSE2609	Information Systems and Databases	CSE1604	2	2	-	1	3		نظم المعلومات وقواعد البيانات
CSE2610	Artificial Intelligence	GEN1601	3	2	-	2	4		الذكاء ألاصطناعي
------	Elective 1 (SR)	------	3	2	2	-	4		اختياري 1 (م.ت)
Total			**17**	**12**	**4**	**7**	**23**		**المجموع**

تدريب ميداني صيفي بالمواقع المتخصصة خارج الكلية لمدة أربعة أسابيع بعد امتحانات الفصل الدراسي الثاني للمستوى الثاني.

Program of Intelligent Systems Engineering (ISE)				برنامج هندسة النظم الذكية					
Level 3 – Semester 1				المستوى الثالث – الفصل الدراسي الأول					
Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
ECE3601	Computer Networks	ECE2606	3	2	-	2	4		شبكات الحاسبات
CSE3602	Digital Control Systems	CSE2608	3	2	-	2	4		أنظمة التحكم الرقمي
CSE3603	Real Time Systems Computer Interfacing	CSE2604	3	2	2	-	4		تقابل الحاسب مع أنظمة الزمن الحقيقي
POW3604	Electrical Power Systems	None	2	2	-	-	2		نظم القوى الكهربية
------	Elective 2 (SR)	------	3	2	2	-	4		اختياري 2 (م.ت)
ECE3605	Graduation Project (1)	None	2	.	-	4	4		مشروع تخرج (1)
Total			**16**	**10**	**4**	**8**	**22**	**المجموع**	

Program of Intelligent Systems Engineering (ISE)				برنامج هندسة النظم الذكية					
Level 3 – Semester 2				المستوى الثالث - الفصل الدراسي الثانى					
Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
POW3606	Programmable Logic Controllers	CSE3602	2	1	-	2	3		ألمتحكمات المنطقية القابلة للبرمجة
ECE3607	Internet of Things	ECE1607	3	2	2	-	4		إنترنت الاشياء
MEC3608	Robotics Technology	None	3	2	-	2	4		تكنولوجيا الروبوتات
ECE3609	Embedded Systems Communications	ECE2607	2	2	-	2	4		اتصالات الأنظمة المدمجة
ECE3610	Wireless Communications	ECE2606	3	2	2	-	4		الاتصالات اللاسلكية
ECE3611	Graduation Project (2)	ECE3605	2	-	-	4	4		مشروع تخرج (2)
Total			**15**	**9**	**4**	**10**	**23**		**المجموع**

Program of Intelligent Systems Engineering (ISE)				برنامج هندسة النظم الذكية					
Level 4 – Semester 1				المستوى الرابع – الفصل الدراسي الأول					
Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
ECE4601	Systems and Networks Security	ECE3610	2	2	-	-	2		تأمين النظم والشبكات
ECE4602	Intelligent Systems	CSE2610	2	1	-	2	3		النظم الذكية
POW4603	Smart Electric Power Networks	POW3604	3	2	2	-	4		شبكات القوى الكهربية الذكية
ECE4604	Mobile Communications	ECE 3601	3	2	-	1	3		الأتصالات المتنقلة
CSE4605	Robotic Systems Programming	MEC3608	2	1	-	2	3		برمجة أنظمة الروبوت
------	Elective 3 (SR)	------	3	2	2	-	4		اختياري 3 (م.ت)
------	Elective 4 (SR)	------	3	2	2	-	4		اختياري 4 (م.ت)
Total			**18**	**12**	**6**	**5**	**23**		**المجموع**

مقررات اختيارية لمتطلبات الجامعة

Elective Courses for University Requirements

(Select 2 courses (4 hrs.))

Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
GEN1008	Report Writing	None	2	2	-	-	2		كتابة التقارير
GEN1009	Problem Solving & Decision-making Skills	None	2	2	-	-	2		مهارات حل المشكلات واتخاذ القرار
GEN1010	Introduction to Marketing	None	2	2	-	-	2		مقدمة في التسويق
GEN1011	Introduction to Economics	None	2	2	-	-	2		مقدمة في الاقتصاد
GEN1012	Innovation and Entrepreneurship	None	2	2	-	-	2		الابتكار وريادة الأعمال
GEN1013	Italian Language	None	2	2	-	-	2		اللغة الايطالية

مقررات اختيارية لمتطلبات المجال

Elective Courses for Discipline Requirements

(Select 2 courses (6 hrs.))

Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
ECE1609	Industrial Electronics	ECE1603	3	2	2	-	4		إلكترونيات صناعية
POW1610	Power Electronics	ECE1603	3	2	2	-	4		إلكترونيات القوى
ECE1611	Laser and its Applications	ECE1603	3	2	2	-	4		الليزر وتطبيقاته
CSE1612	Computer Architecture and organization	GEN0006	3	2	2	-	4		هيكلة وتنظيم الحاسب
ECE1613	Fiber Optics Communications Technology	ECE1606	3	2	2	-	4		تكنولوجيا الاتصالات بالألياف الضوئية
ECE1614	Antennas	ECE1608	3	2	2	-	4		هوائيات

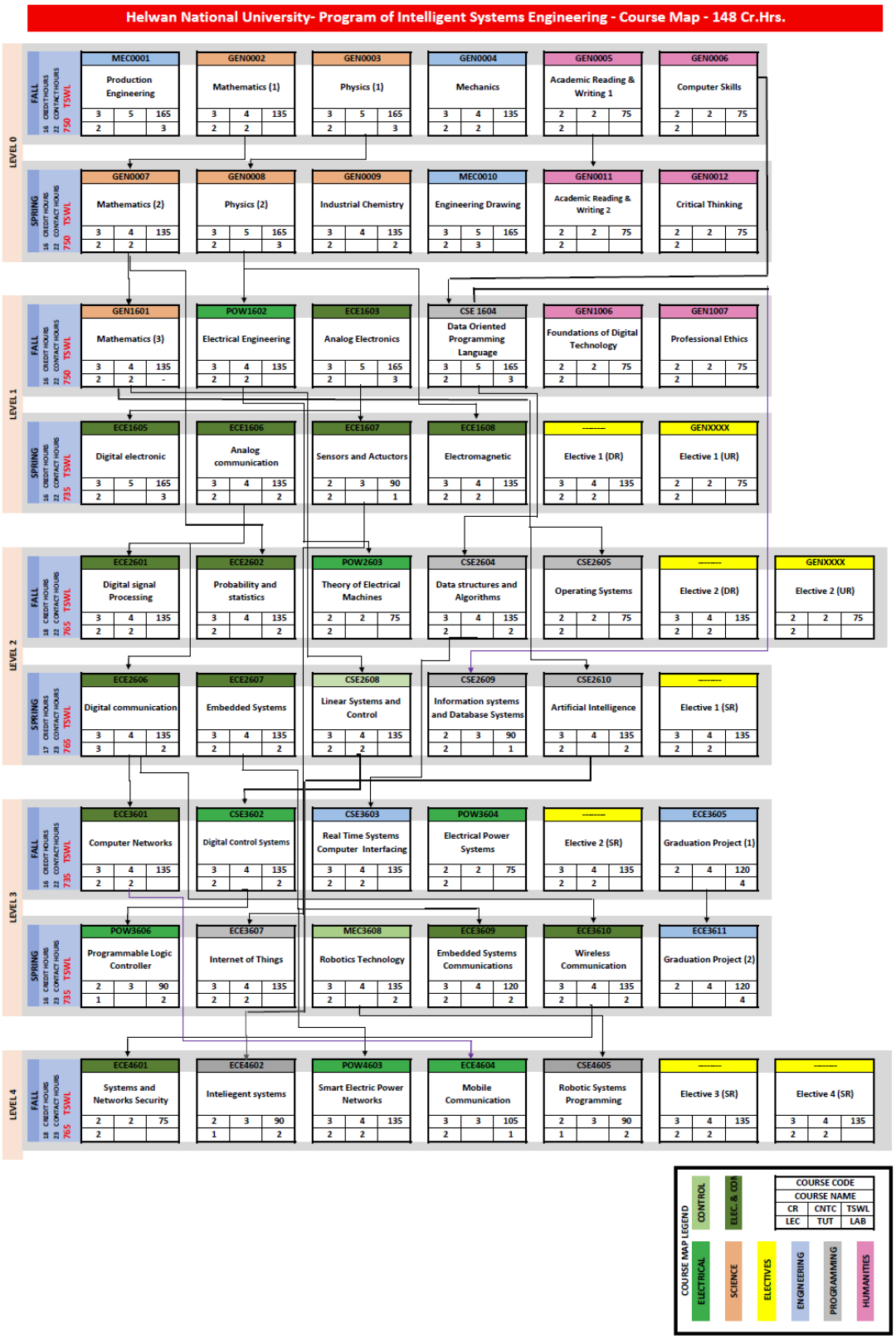
مقررات اختيارية لمتطلبات التخصص

Elective Courses for Specialty Requirements

(Select 4 courses (12 hrs.))

Course Code	Course Name	Prerequisites	Cr. Hrs.	Contact Hours				ECTS	المقرر الدراسي
				Lec	Tut/ Std	Lab/ WS	Total		
POW2611	Systems and Applications of SCADA	POW3604	3	2	2	-	4		أنظمة وتطبيقات SCADA
POW2612	Protection of Electric Power Systems	POW3604	3	2	2	-	4		حماية نظم القوى الكهربية
POW2613	Rationalization and Management of Electric Energy	POW3604	3	2	2	-	4		ترشيد وإدارة الطاقة الكهربية
POW2614	Renewable Energy Systems	POW3604	3	2	2	-	4		منظومات الطاقة المتجددة
POW2615	Analysis of Electrical Power Systems	POW3604	3	2	2	-	4		تحليل نظم القوى الكهربية
POW2616	Energy Distribution Systems	POW3604	3	2	2	-	4		أنظمة توزيع الطاقة
ECE2617	Smart Home Systems	ECE2607	3	2	2	-	4		أنظمة المنازل الذكية
ECE2618	Audio Signals Processing	ECE2601	3	2	2	-	4		معالجة إشارات الصوت
CSE2619	Software Engineering	None	3	2	2	-	4		هندسة البرمجيات
CSE2620	Web Design	CSE1604	3	2	2	-	4		تصميم المواقع الالكترونية على الانترنت
CSE2621	Digital Image Processing	ECE2601	3	2	2	-	4		معالجة الصور الرقمية
CSE2622	Mobile Devices Programming	ECE2604	3	2	2	-	4		برمجة الاجهزة المحمولة
CSE2623	Data Mining	CSE2609	3	2	2	-	4		تنقيب في البيانات
CSE2624	Machine Learning	CSE2610	3	2	2	-	4		تعليم الالة
CSE2625	Parallel and Distributed Systems	CSE2604	3	2	2	-	4		الأنظمة المتوازية والموزعة

ECE3612	Wireless Sensors Networks	ECE3601	3	2	2	-	4		شبكات المجسات اللاسلكية
ECE3613	Smart Cities	ECE3607	3	2	2	-	4		المدن الذكية


Helwan National University- Program of Intelligent Systems Engineering - Course Map - 148 Cr.Hrs.
LEVEL 0
FALL
16 CREDIT HOURS
22 CONTACT HOURS
750 TSWL
MEC0001 Production Engineering 3 5 165 / 2 3
GEN0002 Mathematics (1) 3 4 135 / 2 2
GEN0003 Physics (1) 3 5 165 / 2 3
GEN0004 Mechanics 3 4 135 / 2 2
GEN0005 Academic Reading & Writing 1 2 2 75 / 2
GEN0006 Computer Skills 2 2 75 / 2
SPRING
16 CREDIT HOURS
22 CONTACT HOURS
750 TSWL
GEN0007 Mathematics (2) 3 4 135 / 2 2
GEN0008 Physics (2) 3 5 165 / 2 3
GEN0009 Industrial Chemistry 3 4 135 / 2 2
MEC0010 Engineering Drawing 3 5 165 / 2 3
GEN0011 Academic Reading & Writing 2 2 2 75 / 2
GEN0012 Critical Thinking 2 2 75 / 2
LEVEL 1
FALL
16 CREDIT HOURS
22 CONTACT HOURS
750 TSWL
GEN1601 Mathematics (3) 3 4 135 / 2 2 -
POW1602 Electrical Engineering 3 4 135 / 2 2
ECE1603 Analog Electronics 3 5 165 / 2 3
CSE 1604 Data Oriented Programming Language 3 5 165 / 2 3
GEN1006 Foundations of Digital Technology 2 2 75 / 2
GEN1007 Professional Ethics 2 2 75 / 2
SPRING
16 CREDIT HOURS
22 CONTACT HOURS
735 TSWL
ECE1605 Digital electronic 3 5 165 / 2 3
ECE1606 Analog communication 3 4 135 / 2 2
ECE1607 Sensors and Actuctors 2 3 90 / 2 1
ECE1608 Electromagnetic 3 4 135 / 2 2
------ Elective 1 (DR) 3 4 135 / 2 2
GENXXXX Elective 1 (UR) 2 2 75 / 2
LEVEL 2
FALL
18 CREDIT HOURS
22 CONTACT HOURS
765 TSWL
ECE2601 Digital signal Processing 3 4 135 / 2 2
ECE2602 Probability and statistics 3 4 135 / 2 2
POW2603 Theory of Electrical Machines 2 2 75 / 2
CSE2604 Data structures and Algorithms 3 4 135 / 2
CSE2605 Operating Systems 2 2 75 / 2
------ Elective 2 (DR) 3 4 135 / 2 2
GENXXXX Elective 2 (UR) 2 2 75 / 2
SPRING
17 CREDIT HOURS
23 CONTACT HOURS
765 TSWL
ECE2606 Digital communication 3 4 135 / 3 2
ECE2607 Embedded Systems 3 4 135 / 2 2
CSE2608 Linear Systems and Control 3 4 135 / 2 2
CSE2609 Information systems and Database Systems 2 3 90 / 2 1
CSE2610 Artificial Intelligence 3 4 135 / 2 2
------ Elective 1 (SR) 3 4 135 / 2 2
LEVEL 3
FALL
16 CREDIT HOURS
22 CONTACT HOURS
735 TSWL
ECE3601 Computer Networks 3 4 135 / 2 2
CSE3602 Digital Control Systems 3 4 135 / 2 2
CSE3603 Real Time Systems Computer Interfacing 3 4 135 / 2 2
POW3604 Electrical Power Systems 2 2 75 / 2
------ Elective 2 (SR) 3 4 135 / 2 2
ECE3605 Graduation Project (1) 2 4 120 / 4
SPRING
16 CREDIT HOURS
23 CONTACT HOURS
735 TSWL
POW3606 Programmable Logic Controller 2 3 90 / 1 2
ECE3607 Internet of Things 3 4 135 / 2 2
MEC3608 Robotics Technology 3 4 135 / 2 2
ECE3609 Embedded Systems Communications 3 4 120 / 2 2
ECE3610 Wireless Communication 3 4 135 / 2 2
ECE3611 Graduation Project (2) 2 4 120 / 4
LEVEL 4
FALL
18 CREDIT HOURS
23 CONTACT HOURS
765 TSWL
ECE4601 Systems and Networks Security 2 2 75 / 2
ECE4602 Inteliegent systems 2 3 90 / 1 2
POW4603 Smart Electric Power Networks 3 4 135 / 2 2
ECE4604 Mobile Communication 3 3 105 / 2 1
CSE4605 Robotic Systems Programming 2 3 90 / 1 2
------ Elective 3 (SR) 3 4 135 / 2 2
------ Elective 4 (SR) 3 4 135 / 2 2
COURSE MAP LEGEND
CONTROL
ELECC & COI
ELECTRICAL
SCIENCE
ELECTIVES
ENGINEERING
PROGRAMMING
HUMANITIES
COURSE CODE
COURSE NAME
CR CNTC TSWL
LEC TUT LAB

برنامج هندسة النظم الذكية

Program of Intelligent Systems Engineering

المحتوى العلمى للمقررات الدراسية

Scientific Content of Courses

المحتوى العلمى للمقررات الدراسية

المستوى (0)

Scientific Content of Courses

Level (0)

Course Code	Course Name	Prerequisites	Cr.Hrs	Contact Hours					ECTS	المقرر الدراسي
				Lectures	Tutorial /Studio	LAB	Work shop	Total		
MEC0001	Production Engineering	None	3	2	-	-	3	5		هندسة انتاج
	Course Content: Engineering Materials (Ferrous, Non-Ferrous, Ceramics, ...), Casting Processes (Sand casting, ...), Forming Processes (Forging, Rolling, Drawing ,...), Joining (Riveting, welding,...), Cutting Processes (Turning, Shaping, …), Measuring tools. **Ref.** 1. Swadesh Kumar, "Textbook on Production Technology", Made easy Publications, 2021 2. Andrew Y. C. Nee, "Handbook of manufacturing Engineering and Technology", Springer Link, 2015 3. Kapil Gupta, "Advanced Manufacturing Technologies: Modern Machining, Advanced Joining, Sustainable Manufacturing", Springer-Verlag London, 2017.									
GEN0002	Mathematics (1)	None	3	2	2			4		رياضيات (1)
	Course Content: • Elementary logic. Sets functions. Transformations on graphics. Compositions of functions; inverse functions. • Limits and continuity, Calculus of limits. Discontinuities. Asymptotic. Sequences. Landau symbols. Basic results on limits and on global properties of continuous functions. • Derivatives and derivation rules. Second derivatives and convexity. Differential calculus results (Fermat, Rolle, Lagrange, Cauchy. D L'Hopital Theorems). Taylor approximations. • Primitives and definite integrals. Integration rules. Improper integrals. **Ref.** 1. Erwin Kreyszig, "Advanced Engineering Mathematics", 9th Ed., John Wiley & Sons, 2020. 2. Dennis G. Zill, "Advanced Engineering Mathematics", 7th Ed., Jones & Bartlett Learning, 2020.									
GEN0003	Physics (1)	None	3	2		3		5		فيزياء (1)
	Course Content: Measurements: units and standards, Base units, Vector analysis Kinematics of a particle: Motion along a straight line, Motion in two or three dimensions.Dynamics of a particle, System of particles, Mechanics of rigid body, Elements o fluid mechanics, Thermodynamics. **Ref.** 1- John D. Cutnell, "Introduction to Physics", John Wiely and Sons, 2011 2- Md Nazoor, Simanchala Pinagrahi, "Principles of EngineeringPhysics", Cambridge University Press; 1st edition, 2017									
GEN0004	Mechanics	None	3	2	2			4		ميكانيكا
	Course Content: Statics – Vector application – Equivalent group of forces - Balance of a set of bodies – Vector algebra – Frictions – Moment of inertia -Multiplication of moments of inertia – Theories of moment of inertia transfer - Distributed loads – Body kinematics – Axis –linear motion and motion on a curve- relative motion – Newton									

Course Code	Course Name	Prerequisites	Cr.Hrs	Contact Hours					ECTS	المقرر الدراسي
				Lectures	Tutorial /Studio	LAB	Work shop	Total		
	rules for motion – constant and variable acceleration – applications – work and energy of a body – energy of motion –different types of energy – power. **Ref.** 1. Hibbeller R.C and Gupta A., "Engineering Mechanics: Statics and Dynamics", 14th Edition, Pearson Education , 2015 2. Beer F., Johnston E. , "Vector Mechanics for Engineers: Statics and Dynamics", 12th Edition, McGraw-Hill, 2018.									
GEN0005	**Academic Reading and Writing (1)**	**None**	**2**	**2**				**2**		**القراءة والكتابة الاكاديمية (1)**
	Course Content: The study of Scientific English will focus on (The structure of the four sentence forms in English, simple and progressive, Past, present perfect and perfect tense, The future, imperative, infinitive, gerund, participle, Interrogative and relative pronouns and , Modals: capacity, probability, possibility, duty, necessity, obligation, deduction, Subordinate clauses, Special Constructions,..). The common medical and engineering terms in practice are declared using medical and engineering topics and journal articles. Report writing and abstract writing using accurate grammar and vocabulary are to be focused upon. **Ref.** Teresa Thonney, Academic Writing with Readings: Concepts and Connections, Oxford University Press, 2015.									
GEN0006	**Computer Skills**	**None**	**2**	**2**				**2**		**مهارات الحاسب الآلى**
	Course Content: This course gives a brief introduction to the architecture of the computer as CPU, storage devices, output and input devices, different types of drives. It Introduces students to the concepts and terminologies of problem solving, data representation, Boolean algebra. Also, introduction to programming languages which includes (Input/Output, Conditional Statements, Itrative statements, Arrays). It practices some computer applications in the laboratories such as Visual studio, etc... **Ref.** Jack P. Hoggatt, Jon A. Shank, James R. Smith, Century 21 Computer Skills and Applications, 11th Ed., Cengage Learning, 2020.									
GEN0007	**Mathematics (2)**	**GEN0002**	**3**	**2**	**2**	**-**	**-**	**4**		**رياضيات (2)**
	Course Content: Complex numbers, Polynomials, Vectors: sum, scalar product, wedge product; Vector spaces; subspaces, Bases and dimension, Matrices: elementary operations and properties, Linear Systems: reduction method, Rouché-Capelli theorem, Linear applications and associated matrices, Eigenvalues and eigenvectors, Diagonalization of matrices, Elements of analytic geometry in R^2 and R^3 .									

Course Code	Course Name	Prerequisites	Cr.Hrs	Contact Hours					ECTS	المقرر الدراسي
				Lectures	Tutorial /Studio	LAB	Work shop	Total		
	Ref. 1. Erwin Kreyszig, "Advanced Engineering Mathematics", 9th Ed., John Wiley & Sons, 2020. 2. Dennis G. Zill, "Advanced Engineering Mathematics", 7th Ed., Jones & Bartlett Learning, 2020.									
GEN0008	**Physics (2)**	**GEN0003**	**3**	**2**	**-**	**3**	**-**	**5**		**فيزياء (2)**
	Course Content: Electric field. Electric dipole - electric capacity and electric field energy • Continuous currents: Ohm law - electric resistance • Static magnetic field: magnetic field in vacuum- magnetic induction – Biot Savart law - Laplace law - Lorentz force - Ampere law - Hall effect • Time dependent electric and magnetic fields: electromotive forces and induced currents- induction - energy of magnetic field – Maxwell equations-propagation of electromagnetic waves- Poynting vector - Geometrical optics **Ref.** 1- John D. Cutnell, "Introduction to Physics", John Wiely and Sons, 2011 2- Md Nazoor, Simanchala Pinagrahi, "Principles of EngineeringPhysics", Cambridge University Press; 1st edition, 2017									
GEN0009	**Industrial Chemistry**	**None**	**3**	**2**	**-**	**2**	**-**	**4**		**كيمياء صناعية**
	Course Content: The course has been designed in order to provide students with basic concepts and methods for the understanding and interpretation of chemical processes, of the relationships between structure and properties of materials, from metals to plastics, of the behavior of systems for generation of energy. In order to follow successfully the course, the students are required to possess some introductory chemical notions as chemical nomenclature, concept of atomic and molecular mass, balancing of simple chemical equations. **Ref.** 1- John A. Tyrell, "Fundamentals of Industrial Chemistry: Pharmaceuticals, Polymers, and Business", Wiely, 2014 2- Giora Agam, " Industrial Chemicals: Their Characteristics and Development", Elsevier Science, 2012									
GEN0010	**Engineering Drawing**	**None**	**3**	**2**	**3**	**-**	**-**	**5**		**رسم هندسي**
	Course Content: Drawing tools, Basics of drawing, Lines and numbers, Tangent operations, Projection types, perpendicular projection, isometrics, dimensions, cross-sections, intersections. **Ref.** 1. Hibbeller R.C and Gupta A., "Engineering Mechanics: Statics and Dynamics", 14th Edition, Pearson Education , 2015 2. Beer F., Johnston E. , "Vector Mechanics for Engineers: Statics and Dynamics", 12th Edition, McGraw-Hill, 2018.									

Course Code	Course Name	Prerequisites	Cr.Hrs	Contact Hours					ECTS	المقرر الدراسي
				Lectures	Tutorial /Studio	LAB	Work shop	Total		
GEN0011	Academic Reading and Writing (2)	GEN0005	2	2				2		القراءة والكتابة الاكاديمية (2)
	Course Content: The course mainly focuses on advanced persuasive writing skills. How to analyze argumentative/persuasive texts. How to evaluate the strength of an argument. How to use and cite sources in their writing. How to write a persuasive opinion editorial/position paper. **Ref.** Teresa Thonney, Academic Writing with Readings: Concepts and Connections, Oxford University Press, 2016.									
GEN0012	Critical Thinking	None	2	2				2		التفكير النقدى
	Course Content: It designed to help freshman and sophomore students learn key cognitive skills, attention to which will aid their success in all their course work and critically analyzing published research articles. **Ref.** Katherine H. Pherson and Randolph H. Pherson, Critical Thinking for Strategic Intelligence, Sage Publications Inc, 2020.									

المحتوى العلمى للمقررات الدراسية

المستوى (1)

Scientific Content of Courses

Level (1)

GEN1601	**Mathematics (3)**	GEN0007	**3**	**2**	**2**			**4**		**رياضيات (3)**
	Course Content: • Numerical Series; series of functions. Taylor and power series. Fourier series. • Real functions of several variables: limits, continuity and differentiation. •Multiple integrals (Linear, douple, triple) •Computation of the area, and volume •Differential equations: separable variables, first and second order linear equations, linear differential systems with constant coefficients •Complex Analysis: Analytic functions - Fourier transform - Laplace transform. **Ref.** 1. Erwin Kreyszig, "Advanced Engineering Mathematics", 9th Ed., John Wiley & Sons, 2020. 2. Dennis G. Zill, "Advanced Engineering Mathematics", 7th Ed., Jones & Bartlett Learning, 2020.									
POW1602	**Electrical Engineering**	**GEN0008**	**3**	**2**	**2**	**-**	**-**	**4**		**هندسة كهربية**
	Course Content: Basic laws and definitions – Resistors – Inductors – Capacitors - Active elements of electric circuits - Kirchhoff's law - Methods of solution of D.C. circuits - Fundamentals of A.C. circuits - Logic circuits – Transformers - Equivalent circuits for transformer - Generators and motors - Induction motor - D. C. motors - Stepper motors. **Ref.** 1. S. Bobby Rauf, "Electrical Engineering Fundamentals", CRC Press, 2020. 2. Thaddeus Roppel, Charles A. Gross, "Fundamentals of electrical engineering", CRC Press, 2012.									
ECE1603	**Analog Electronics**	**None**	**3**	**2**	**-**	**3**	**-**	**5**		**الكترونيات تناظرية**
	Course Content: Diodes, The diode as a circuit element, Analysis of diode circuits. Zener diodes, special diode, diode applications. • Bipolar junction transistors (BJT), characteristics, Biasing. DC and AC models, small signal amplifiers. Field effect transistors (FETs), types, characteristics, small signal models. Operational amplifier Op-Amp characteristics and analysis. Op-Amp applications. Power amplifiers. Oscillators, multi vibrators. **Ref.** 1. Adel S. Sedra, Kenneth C. Smith, Tonny Chan Carusone, Vincent Gaudet , "Microelectronic Circuits", Oxford University Press, 8th Ed., 2020. 2. Robert Boylestad, Louis Nashelsky, "Electronic devices and circuit theory", Pearson, 2014.									
CSE1604	**Data Oriented Programming Language**	GEN0006	**3**	**2**	**-**	**3**	**-**	**5**		**لغات برمجة بيانات**
	Course Content: Learn functional programming that's include Functions and Recursive functions, Pointers, Strings, Files, and Structures. Also, introduce concepts of algorithms and data structure. Learn Object Oriented Programming Concepts which includes (Objects and Classes, Inheritance, Polmerphism, Java class Hierarchy, Files, Exception handling, Collection, GUI applications and overview about concurrent programming and synchronization). Finally, learn concepts of Data oriented Programming and									

	distinguish data-oriented programming from functional and Object Oriented Programming . **Ref.** 1. E. Balagurusamy, "Object Oriented Programming and Data Structures", McGraw-Hill Education, 2014. 2. Yehonathan Sharvit, "Data Oriented Programming: Unlearning Objects", Version 2-Manning Publications, 2021.									
GEN1006	**Foundations of Digital Technology**	**None**	2	2				2		**اسس تكنولوجيا رقمية**
	Course Content: This foundation course in Digital Technologies is designed to prepare you with the knowledge and professional skills needed for progression onto your degree. It prepares students to understand and apply technological concepts and processes that are the cornerstone for the high school technology program. Group and individual activities engage students in creating ideas, developing innovations, and engineering practical solutions. **Ref.** 1. Anant Agarwal and Jeffrey Lang, Foundations of Analog and Digital Electronic Circuits (The Morgan Kaufmann Series in Computer Architecture and Design), Morgan Kaufmann, 2005.									
GEN1007	**Professional Ethics**	None	2	2				2		**اخلاقيات المهنة**
	Course Content: Students learn about the principles that govern the behavior of a person or group in a business environment. Like values, professional ethics provide rules on how a person should act towards other people and institutions in such an environment. **Ref.** 1. Charles E. Harris, Jr., Michael S. Pritchard, Michael J. Rabins, Ray James and Elaine Englehardt, Engineering Ethics: Concepts and Cases , 5th Ed, Cengage Learning, 2014.									
ECE1605	**Digital Electronics**	**ECE1603**	**3**	**2**		**3**	**-**	**5**		**الكترونيات رقمية**
	Course Content: Introduction, Number Systems, Binary Arithmetic, and Codes. Boolean algebra and Logic Gates, The Karnaugh Map and Logic Simplifications, Design of RTL, Combinational Logic Circuits, Half Adders and Full Adders, Comparators, Encoders and Decoders, Code Conversions, Multiplexers and De-multiplexers, Latches and Flip-Flops, Counters, Shift Registers, Memory Circuits, Programmable logic devices (PLDs), Arithmetic and Logic Unit, Hardware Description Language (Verilog), Synchronus sequential logic. **Ref.** 1. Roger L. Tokheim, Patrick E. Hoppe, "Digital Electronics: Principles and Applications", McGraw-Hill Education, 2021. 2. M. Morris R. Mano, Michael D. Ciletti, " Digital Design With an Introduction to the Verilog HDL, VHDL, and SystemVerilog", Pearson, 2017.									

ECE1606	**Analog Communications**	None	**3**	**2**		**2**	**-**	**4**		**إتصالات تناظرية**
	Course Content: the fundamental concepts of analog modulation systems, signals and spectra, electromagnetic spectrum and its usage, communication channels and propagation characteristics, amplitude modulation and demodulation - spectra, circuits and systems, frequency modulation/demodulation, frequency division multiplexing, radio transmitters and receivers, spectra, circuits & systems. **Ref.** 1. Zhi Ding, Bhagwandas Pannalal Lathi, "Modern digital and analog communication systems", Oxford University Press, USA, 5th Ed., 2019 2. Leon W. Couch , "Digital and Analog Communication Systems:, Prentice Hall, 8th ed., 2012.									
ECE1607	**Sensors and Actuators**	ECE 1603	**2**	**1**		**2**	**-**	**3**		**المستشعرات و المفعلات**
	Course Content: Transducers, sensors and actuators: Types of transducers, types of sensors and actuators. Sensor Characteristics: Transfer functions, accuracy, precision, sensitivity, hysteresis, systematic and random errors, range and repeatability. Sensor and Actuator Design and Modelling: Physical principles of sensing: Charge, Capacitance, Induction, Resistance, Piezoelectric effect, hall effect. Actuator control: DC motor, servo motor and stepper motor control. Sensor and Actuator applications: Sensor and actuator applications in e- health, robotics. Measurements and Error: Fundamentals of Measurement Theory, Performance Characteristics of Instruments, Measurement Errors, Statistical Analysis of Measurements. Data Acquisition Systems: Signal Conditioning, Analog to Digital Conversion, Digital to Analog Conversion. DC Analog Instruments: Deflection-type Instruments, DC Electronic Voltmeter, Resistance Measurement. AC Analog Instruments: Rectifier-type Instruments, Thermocouple Instruments, Frequency and Phase Measurements: Introduction to oscilloscope, Architecture of oscilloscope, Frequency Measurement, Phase Measurement. Digital Instruments: Digital Voltmeter Systems, Counting Circuits, Digital Frequency Meter, Digital Millimeters. **Ref.** 1. Francisco Andre Correa Alegria, "Sensors And Actuators", World Scientific Publishing Co Pte Ltd, 2021. 2. Nathan Ida, "Sensors, Actuators, and Their Interfaces: A multidisciplinary introduction", The Institution of Engineering and Technology, 2nd ed., 2020.									
ECE1608	**Electromagnetic**	**GEN0008**	**3**	**2**	**2**	**-**	**-**	**4**		**كهرومغناطيسية**
	Course Content: Time varying fields and Maxwell's equations, uniform plane wave propagation in free space media, uniform plane wave propagation in lossy dielectric media, wave polarization, reflection and transmission of plane wave at normal and oblique incidence, parallel and perpendicular polarization, Brewster's and critical angles. Lossy and lossless transmission line analysis, reflection coefficient, power transmitted, matching techniques, Smith chart, coaxial and micro-strip lines, signal propagation along dispersive lines. Rectangular and circular waveguides, cavity resonators. **Ref.** 1. William H. Hayt, Jr. and John A. Buck, "Engineering Electromagnetics", McGraw-Hill Education, 9th ed., 2018. 2. Nathan Ida, "Engineering Electromagnetics", 4th ed, Springer., 2021.									

المحتوى العلمى للمقررات الدراسية

المستوى (2)

Scientific Content of Courses

Level (2)

ECE2601	**Digital Signal Processing**	ECE1606	**3**	**2**	**2**	**-**	**-**	**4**		**معالجة إشارات رقمية**
	Course Content: Discrete-time signals and systems: Discrete-time signals, Discrete-time systems: memory less systems, linear systems, time invariant systems, causality, stability, linear difference equations, Frequency-domain representation of discrete-time signals and systems. The Z- transform. Discrete Fourier transform: Discrete Fourier series - properties of DFS - periodic convolution– linear convolution using DFT - computation of DFT - circular convolution - decimation in time and decimation in frequency algorithms - FFT algorithm for a composite number. IIR and FIR Filter structures: Signal flow graph representation - basic filter structures - structures for linear phase – finite word - length effects in digital filters - quantizer characteristics - Digital filter design: Design of IIR digital filters from analog filters - Butterworth and Chebyshev filters –impulse invariant and bilinear transformation methods - spectral transformation of IIR filters - FIR filter design - linear phase characteristics - window method. General and special purpose hardware for DSP **Ref.** 1. Anastasia Veloni, Nikolaos I. Miridakis, Erysso Boukouvala, "Digital and Statistical Signal Processing: CRC, 2019. 2. Juan Zhang, "Digital Signal Processing: Fundamentals, Techniques and Applications". Nova Science Pub Inc, 2016. 3. Samir I. Abood, "Digital Signal Processing: A Primer with MATLAB", CRC Press, 2020.									
ECE2602	**Probability and statistics**	**GEN0007**	**3**	**2**	**2**			**4**		**الاحتمالات والاحصاء**
	Course Content: Sample Space, Set Theory and Operations, Definition of Probability, Counting Techniques, Conditional Probability, Independence and Bayes Theorem, Discrete Random Variables, Continuous Random Variables, Cumulative Distribution Function, Expectation and Variance, Joint Probability Function, Conditional Probability Function, Conditional Expectation, Some Discrete Distributions, Some Continuous Distributions, Graphical Methods for Describing, Describing Data with Numbers, Statistical Inference, Confidence Intervals for One Sample , Confidence Intervals for Two Samples , Hypothesis Testing - One Sample , Hypothesis Testing - Two Samples , Linear Regression, Regression Assumptions **Ref.** Narayan C. Giri, "Introduction to Probability and Statistics", 2019. E. Rukmangadachari, "Probability and Statistic", 2010.									
POW2603	**Theory of Electrical Machines**	**POW1602**	**2**	**2**	**-**	**-**	**-**	**2**		**نظرية آلات كهربية**
	Course Content: Theory and Operation of Transformers, Energy Conversion, DC Machines: Theory and Operation, Field Excitation, Speed Control, AC Machines: Theory and Operation, Induction Machines, Speed Control, Simulation if Electric Machines, Special Purpose Machines.									

	Ref. 1. Sahdev, S. K, "Electrical machines", Cambridge University Press, 2018. 2. S. K. Bhattacharya, "Electrical machines", McGraw-Hill Education, 2015.									
CSE2604	**Data Structures and Algorithms**	**CSE1604**	**3**	**2**	**-**	**2**	**-**	**4**		**هياكل البيانات والخوارزميات**
	Course Content: In this course, use and analyze different data structures techniques, such as Array, linked lists, Stacks, Queue, Trees, Binary search tree, Dictionaries, Graph visit and its applications (Shortest Path and Minimum Spanning Tree), and Hashtables. Also student study sorting which includes: Quadratic algorithms like (Bubble, Insertion, and Selection sort, etc), Logarithemic algorithm like (Heap and Quick sort, etc)) and Linear sort like Counting sort, etc. Also, learn Search based algorithms. Finally, Learn how to calculate the complexity analysis of recursive and non recursive algorithms by appling asymptotic Big-O analysis to describe the performance of algorithms and evaluate which strategy to use for efficient data retrieval. **Ref.** 1. Timothy A. Budd, "An Introduction to Object-Oriented Programming", Pearson, 2021. 2. Duane A. Bailey , " Java Structures: Data Structures in Java for the Principled Programmer ", Mcgraw-Hill, 2007.									
CSE2605	**Operating Systems**	**CSE1604**				**2**		**2**		**-**
	Course Content: Acquire the basic operating system concepts such as processes & threads. Also, explain CPU scheduling, Pthreads, Process management, synchronization and Deadlocks. Pthreads in java.. Explain the Details of memory management (segmentation and paging), I/O, and File systems. Illustrate the above concepts on existing operating systems such as UNIX/LINUX and Windows a case studies. **Ref.** 1. Abraham Silberschatz, Peter B. Galvin , Peter B. Galvin "Operating System Concepts", 9th Edition, Wiley, 2013.									
ECE2606	**Digital Communications**	**ECE1606**	**3**	**2**		**2**	**-**	**4**		**الاتصالات الرقمية**
	Course Content: The basic concepts of Digital Communication, base band, pass band modulation and to give an exposure to error control coding and finally to discuss about the spread spectrum modulation schemes. **Ref.** 1. Zhi Ding, Bhagwandas Pannalal Lathi, "Modern digital and analog communication systems", 5th Ed , Oxford University Press, USA,., 2019 2. Leon W. Couch, "Digital and Analog Communication Systems:, 8th Ed., Prentice Hall, 2012.									

ECE2607	**Embedded Systems**	**None**	**3**	**2**	**-**	**2**	**-**	**4**		**الأنظمة المدمجة**
	Course Content: Computer systems, constraints present in typical embedded systems, typical components of an embedded system, embedded system classification. Embedded System Requirements and Specifications: Requirements documents, functional and non-functional requirements, specification documents and verification plans, executable specifications. Embedded System Modeling: Modeling based on first principles, state machines, control and dataflow diagrams, Fourier, Laplace and z transforms. Instruments used in verification, waveforms generators, oscilloscopes, multi-meters, logic analyzers. Embedded System Implementation Platforms: Architecture and ISA of the microcontroller, GPP and DSP. FPGAs as embedded system implementation platforms. Programming for embedded systems: Analog and Digital Input and Output, programming in high and low-level languages, program optimization debugging using breakpoints, LEDs and logic analyzers. Real-time distributed embedded systems: The real-time environment. Target applications – target design parameters, constraints and challenges (performance, cost, power consumption). Measuring time and clock synchronization. Real-time system modeling: abstractions for real-time embedded system modeling, structural elements of a model, temporal and logical control, worst-case execution time. Real-Time communication: implicit and explicit flow control, requirement analysis for real-time communication, limitations of the OSI model for real-time systems, design of critical real-time communication networks. Real-time operating systems: task soft and hard deadlines, static and dynamic memory management. Embedded system design with FPGAs: Embedded system design with platform FPGAs, HDLs, High-Level Synthesis. **Ref.** 1. K. C. S. Murti, “Design principles for embedded systems”, Springer, 2022. 2. James K. Peckol, “EMBEDDED SYSTEMS: A Contemporary Design Tool”, JOHN WILEY, 2018. 3. Jiacun Wang, “Real-Time Embedded Systems”, Wiley, 2017.									
CSE2608	**Linear Systems and Control**	GEN1601	**3**	**2**	**2**	**-**	**-**	**4**		**الأنظمة الخطية و التحكم**
	Course Content: System Representation (Time Domain & Frequency Domain)-Open Loop and Closed loop Control Systems-Modeling (Electrical Systems- Mechanical Systems- Electromechanical Systems) - Block Diagrams-Transfer Function (Block Diagram Reduction- SFG)-Second Order Systems (Response and Analysis)-Error Analysis (Steady State - Transient)-PID Controller-Error Criteria (ISE- IAE)-Stability Analysis (Routh Hurwitz Criterion) - Nyquist Plot (Analysis – Design) - Root Locus (Analysis - Design)-Bode Plot (Analysis - Design) - Controller Design with constraints -State Space Representation-Solution of State Space Representation-Controllability and Observability. **Ref.** **1.** Norman S. Nise, "Control Systems Engineering", 8th Ed., John Wiley, 2019									

<table>
<tr><td></td><td colspan="10">2. Ogata K., Modern Control Engineering, 5th Ed., Prentice Hall, 2010</td></tr>
<tr><td rowspan="2">CSE2609</td><td>Information Systems and Databases</td><td>CSE1604</td><td>2</td><td>2</td><td>-</td><td>1</td><td>-</td><td>3</td><td></td><td>نظم المعلومات وقواعد البيانات</td></tr>
<tr><td colspan="10">Introduction to information system, Entrprise businese system, System development life cycle, Security of Information Technology, Difference between application software and system software. Basic concepts of database systems and its architecture. Also, includes database management systems (DBMS), data models, database languages, database users and administrators, the design and creation of database systems, models of database management systems, three schema architecture of DBMS. Entity-Relationship Model: Basic Concepts, Design Issues, Mapping Constraints, Keys, Entity-Relationship (ER) Diagram, Weak Entity Sets, Extended ER features. Relational Model: Structures of relational databases, Integrity Constraints, Logical database Design, Tables, Views, Relational Algebra, Mapping ER Diagram to Relational model. Structured Query Language (SQL): Basic Structures, DDL and DML languages, SQL queries and Integrity Constraints. Relational. Using of DBMS systems such as MySQL, SQLserver, Microsoft access DBMS and others.
<u>Ref.</u>

1. Coronel, Carlos, and Steven Morris. "Database systems: design, implementation, & management." Cengage Learning, 2016.
2. Kenneth C. Laudon and Jane P. Laudon.”Management Information Systems: Managing the Digital Firm”, Pearson, 2020.</td></tr>
<tr><td rowspan="2">CSE2610</td><td>Artificial Intelligence</td><td>GEN1601</td><td>3</td><td>2</td><td>-</td><td>2</td><td>-</td><td>4</td><td></td><td>الذكاء ألاصطناعي</td></tr>
<tr><td colspan="10">Course Content:

This course provides a solid foundation in AI concepts, techniques, and applications, equipping you with the skills needed to understand and build intelligent systems. Introduction to Artificial Adaptive systems, Top down producers versus bottom up procedures, Information approach, Theory and types of graphs, supervised and unsupervised learning and validation.

<u>Ref.Ref.</u>

1. Stuart J. Russell, Peter Norvig, “Artificial Intelligence: A Modern Approach”, 3rd Ed., Pearson Education, 2016.
2. Dheeraj Mehrotra, “Basics of Artificial Intelligence & Machine Learning”, Notion, 2019</td></tr>
</table>

المحتوى العلمى للمقررات الدراسية

المستوى (3)

Scientific Content of Courses

Level (3)

ECE3601	**Computer Networks**	**ECE2606**	**3**	**2**		**2**	**-**	**4**		**شبكات الحاسبات**

Course Content:

Introduction to computer networks - Network Topologies - Network Components and Devices - Network Classifications (PAN, LAN, MAN and WAN) - Introduction to Local Area Network (LAN) - IEEE Standards for LAN (802 Project) - Ethernet/802.3 LANs - Ethernet Data link Protocol - Ethernet implementations – Switched Ethernet – Fast and Giga Ethernet LANs - 40Gbps and 100Gbps Ethernet LANs - Token ring and FDDI networks. Virtual LAN (VLAN)–Wireless LAN (WLAN) - Wireless LAN Standards (IEEE 802.11) – IEEE 802.11 WLAN physical layer - WLAN Media Access Control (CSMA/CA) - WLAN Modes and Operation – WLAN Security - Introduction to Routing and Packet Forwarding - Routing Table Structure and Lookup Process - Static and Dynamic Routing – IP Version 6. Distance Vector Routing Protocols - Least cost routing algorithms - Principles of internetworking and interworking devices.

Ref.

1. Larry L. Peterson, Bruce S. Davie, "Computer Networks: A Systems Approach", Morgan Kaufmann, 2021.
2. Andrew S. Tanenbaum, Nick Feamster, David J. Wetherall, "Computer Networks", Pearson, 2021

CSE3602	**Digital Control Systems**	**CSE2608**	**3**	**2**	**-**	**2**	**-**	**4**		**أنظمة التحكم الرقمي**

Course Content:

Introduction to digital control systems, Discrete time systems and review on the z-transform, Sampling and Reconstruction, State variables in discrete time systems, Closed-loop discrete time systems, Sampled-data transformation of analog controllers, Digital controller design; lead-lag and PID controllers, Optimal control design method. State-space representation and control in discrete time, Controllability, observability, pole placement. State-space representation and control in discrete time.

Ref.

1. M. Sami Fadali, "Digital Control Engineering: Analysis and Design", 3rd Ed., Academic Press, 2019.
2. Raymond G Jacquot, "Modern Digital Control Systems", 2nd Ed., CRC Press , 2018.
3. Aranya Chakrabortty, H. Troy Nagle, Charles L Phillips, " Digital Control System Analysis & Design", Global Edition-Pearson Education Limited, 2015

CSE3603	**Real Time Systems Computer Interfacing**	**CSE 2604**	**3**	**2**	**2**	**-**	**-**	**4**	تقابل الحاسب مع أنظمة الزمن الحقيقي

Course Content:

Basic real-time systems concepts, Raspberry Pi architecture, Usage of an oscilloscope, Cross compilation, Response time and latency, GPIO, Multithreading in C++, Socket and interprocess communications, Software PWM, Basic electrical circuits, RMA, Sensors and input devices, Open CV introduction and applications. Buffering and queuing theory, Reliability, performance and ethical concerns, Assessment, review, and current.

Ref.

1. Jane W. S. Liu, "Real Time Systems", Pearson Education Publication, 2020.
2. Mall Rajib, "Real Time Systems" , Pearson Education, 2018
3. Albert M. K. Cheng, "Real-Time Systems: Scheduling, Analysis, and Verification", Wiley, 2020.

POW3604	**Electrical Power Systems**	None	**2**	**2**	**-**	**-**	**-**	**2**		**نظم القوى الكهربية**
	Course Contents: This course is an introductory subject in the field of electric power systems and electrical to mechanical energy conversion. Electric power has become increasingly important as a way of transmitting and transforming energy in industrial, military and transportation uses. Electric power systems are also at the heart of alternative energy systems, including wind and solar electric, geothermal and small scale hydroelectric generation. **Ref.** 1.B. M. Weedy, B. J. Cory, N. Jenkins, Janaka B. Ekanayake, Goran Strbac," Electric Power Systems, 5th Edition ", WIELY, 2012									
ECE3605	**Graduation Project (1)**	None	**2**	**.**	**-**	**4**	**-**	**4**		**مشروع تخرج (1)**
	Course Content: The student introduces a complete study and constructs the basic designs and calculations for the project determined through his department applying the basics of science that he studied - The project includes intelligent systems design, electronics, control and programming – The student presents a report containing the rules and calculations that he made in his project.									
POW3606	**Programmable Logic Controllers**	CSE3602	**2**	**1**	**-**	**2**	**-**	**3**		**ألمتحكمات المنطقية القابلة للبرمجة**
	Course Content: Typical PLC hardware specification, Programming languages LAD, FBD, IL, SFC, ST, Digital Inputs and Outputs, applications. Overview of Analogue Inputs and Outputs, types and applications. Flags and their applications. Further application program development using MELSOFT GX Works2, Simple program search techniques. X-referencing. Timing applications, Counting application. Introduction to special function, Consolidation application program development, Introduction to program documentation, Diagnostic techniques. **Ref.** 1. William Bolton, "Programmable Logic Controllers", 6th Ed., Elsevier Science, 2019. 2. Frank Petruzella, "Programmable Logic Controllers", 5th Ed., McGraw-Hill, 2016.									
ECE3607	**Internet of Things**	ECE 1607	**3**	**2**	**2**	**-**	**-**	**4**		**إنترنت الاشياء**
	Course Content: What Is the Internet of Things? Machine to Machine / User-less Communication, Common Use Cases, Components of an IoT Solution, Open Source and Commercial Examples, Competing Standards for IoT. IoT specialization: Industrial, Medical/Healthcare, Automotive, Energy/Utilities, Financial, Brainstorming									

	IoT Utilization, Acquiring Data, Traditional Data Storage, Analog and Digital I/O Basics, Sensors and Data Collection Points, Embedded Platforms / Microcontrollers, Software Development. Collecting and Storage of IoT Sensor Data, Data Aggregation, Processing IoT Data, Privacy and Security, Analysis and Visualization of Data. Implementing IoT, implementing a Multi-Node IoT Solution, IoT Analytics, IoT Strategies, IoT Governance and Management Strategies, What's Next in IoT, Design an IoT Solution. **Ref.** 1. Sudhir Kumar, "Fundamentals of Internet of Things", Chapman and Hall_CRC, 2021. 2. Abhik Chaudhuri, "Internet of things, for things, and by things", CRC Press, 2019. 3. Jamil Y. Khan, Mehmet R. Yuce, Internet of Things (IoT): Systems and Applications", Pan Stanford Publishing Pte Ltd, 2019.									
MEC3608	**Robotics Technology**	**None**	**3**	**2**	**-**	**2**	**-**	**4**		**تكنولوجيا الروبوتات**
	Course Content: Introduction to robotics – Kinematic modeling: Spatial representations and transformations - homogeneous transformations - Forward and inverse kinematics; Jacobean for velocities and static analysis – Introduction to robot dynamics and control. Fundamentals of parallel robot mechanisms - Design Criteria - Five Degrees of Freedom Parallel Robotic Machine - Planar Parallel Robotic Machine Design - Spatial Parallel Robotic Machines with Prismatic Actuators - Spatial Parallel Robotic Machines with Revolute Actuators - Reconfigurable Parallel Kinematic Machine Tools. **Ref.** 1. Saeed B. Niku, "Introduction to Robotics: Analysis, Control, Applications," Wiley, 2020. 2. Matjaž Mihelj, Tadej Bajd, Aleš Ude, Jadran Lenarčič, Aleš Stanovnik, Marko Munih, Jure Rejc, Sebastjan Šlajpah, "Robotics", Springer International Publishing, 2019. 3. Mordechai Ben-Ari, Francesco Mondada, "Elements of Robotics", Springer, 2018.									
ECE3609	**Embedded Systems Communications**	ECE2607	**2**	**2**	**-**	**2**	**-**	**4**		**اتصالات الأنظمة المدمجة**
	Course Content: Introduction to Microcontrollers options (C2000, Arduino, Raspberry Pi), C2000 Launchpad Architecture and More details, Input/output ports and Peripheral Interfacing, Timers, Interrupts, USB Comm., Input/output devices, Interrupts, Pulse width modulation and other applications, Communication and communication protocols. **Ref.** 1. K. C. S. Murti, "Design principles for embedded systems", Springer, 2022. 2. James K. Peckol, "Embedded Systems: A Contemporary Design Tool", John Wiley, 2018. 3. Jiacun Wang, "Real-Time Embedded Systems", Wiley, 2017.									

ECE3610	**Wireless communications**	ECE2606	**3**	**2**	**2**	**-**	**-**	**4**		**الاتصالات اللاسلكية**
	Course Content: This course introduces fundamental technologies for wireless communications. It covers the characterization of the radio signal, including radio wave propagation, Path loss, slow fading, fast fading, Multipath, diversity techniques and wireless channel modeling. Topics of this course include the multiple access technologies such as TDMA/FDMA, CDMA, OFDMA and NOMA. **Ref.** 1. William C. Y. Lee, "Wireless and Cellular Communications", 3rd. Ed., McGraw-Hill, 2006. 2. K. Daniel Wong - Fundamentals of Wireless Communication Engineering Technologies-Wiley (2012)									
ECE3611	**Graduation Project (2)**	ECE3605	**2**	**.**	**-**	**4**	**-**	**4**		**مشروع تخرج (2)**
	Course Content: The student completes study of the project and constructs the determined design including intelligent systems design, electronics, control and programming – The student presents a report containing the rules and calculations that he made in his project.									

المحتوى العلمى للمقررات الدراسية

المستوى (4)

Scientific Content of Courses

Level (4)

	Systems and Networks Security	ECE3610	**2**	**2**	**-**	**-**	**-**	**2**		**تأمين النظم والشبكات**
ECE4601	**Course Content:** Information Security Services – Security Hacking types – Classical Encryption Schemes- Principles of Modern Cryptography - Fundamentals of Information Security – Symmetric Encryption Algorithms including S-DES, DES, AES - Asymmetric Encryption Algorithms including RSA, Elgamal Encryption – Digital Electronic Signature – Message Authentication-Secrecy vs. Integrity-Message Authentication Codes-CBC-MAC-Authenticated Encryption – Hash Functions-Security Requirements-HMAC-Additional Applications of Hash Functions – Key Distribution and Key Management – Diffie-Hellman Key Exchange – Authentication Certificates – Security in Computer Networks and Information. **Ref.** **1.** William Stallings, Lawrie Brown, "Computer Security_ Principles and Practice", Pearson Education Limited, 2018. **2.** John R. Vacca, "Network and System Security", 2nd Ed., Syngress, 2013. **3.** William Stallings, "Network Security Essentials: Applications and Standards", 4th Ed., Prentice Hall, 2010.									
	Intelligent Systems	CSE2610	**2**	**1**	**-**	**2**	**-**	**3**		**الانظمة الذكية**
ECE4602	**Course Content:** This course explores intelligent systems, definition and types of intelligent systems, artificial intelligence (AI) techniques and their practical applications. Students will explore foundational concepts, including search algorithms, evolutionary algorithms, natural language processing. Decision Support Systems (DSS) and expert systems. Design, implementation, and evaluation of intelligent systems, case studies and real-world Intelligent Systems. **Ref.** 1. Christopher M. Bishop," Pattern Recognition and Machine Learning", Springer, 2006. 2. Stuart Russell and Peter Norvig, " Artificial Intelligence: A Modern Approach", Pearson, 2020 .									
	Smart Electric Power Networks	POW3604	**3**	**2**	**2**			**4**		**شبكات القوى الكهربية الذكية**
POW4603	**Course Content**: Distribution System Fundamentals, Load Characteristics, Transformers – Single Phase & Three-Phase, Substations & Primary Feeders, Overhead Lines & Modeling, Underground Distribution & Modeling, Voltage Regulation, Approximate Voltage Drop Analysis, Power Flow Analysis. Capacitor Applications, Faults/Short-Circuit Analysis, Short-Circuit Protection Operation, Reliability Analysis, Power Quality, Distribution Management Systems, Integrated Volt/Var Control, Automated Fault Location, Isolation and Service Restoration, Load Modeling and Demand Response, Distributed Energy Resources & Microgrids. **Ref.** 1. Chengshan Wang, Jianzhong Wu, Janaka Ekanayake, Nick Jenkins, "Smart Electricity Distribution									

	Networks", CRC Press, 2020. 2. Hossam Gabbar, "Smart Energy Grid Engineering", Academic Press, 2017.									
ECE4604	**Mobile Communication**	ECE3601	**3**	**2**	**-**	1	**-**	**3**		**الأتصالات المتنقلة**
	Course Content: This course is developed to provide principal concepts of Mobile Communications. It covers the cellular systems from GSM to LTE-Advanced with their network architecture, techniques, procedures, channels and protocol stack. The tutorials enhance the understanding of the theoretical part by introducing examples and solving problems. The practical part covers the radio signal fundamentals, characteristics and propagation in different environment and for different modulation schemes. **Ref.** 1- Theodore S. Rappaport, 2002, "Wireless Communication: principles and practice", 2nd Edition, Upper Saddle River, NJ: Prentice Hall, John Wiley and Sons Ltd. 2- Sabih Güzelgöz and Hüseyin Arslan, 2010, "A Wireless Communications Systems Laboratory Course", IEEE Transactions on Education, vol. 53, no. 4									
CSE4605	**Robotic Systems Programming**	MEC3608	**2**	**1**	**-**	**2**	**-**	**3**		**برمجة أنظمة الروبوت**
	Course Content: Robot programming languages, C++ for Robot Programming with simple project, Python for Robot Programming with simple application, JAVA for Robot Programming with simple application, MATLAB for Robot Simulation and Programming. **Ref.** 1. Ashish Dutta, "Robotic Systems - Applications, Control and Programming", InTech, 2012. 2. Ayssam Elkady, Tarek Sobh, "Robotic Systems: Applications, Control and Programming", Magnum Publishing LLC, 2016.									

المحتوى العلمى للمقررات الاختيارية لمتطلبات الجامعة

Contents of Elective Courses for University Requirements

Course Code	Course Name	Prerequisite	Cr. Hours	Contact Hours					ECTS	المقرر الدراسي
				Lec.	*Studio / tutorial*	*Lab*	*Work shop*	total		
GEN1008	**Report Writing**	None	2	2				2		كتابة التقارير
	Course Content: Writing principles (Report organization – Forms - Graphics) - Report contents – Report steps (Report outlines & revising) – Standard abbreviations - Letters - Page design - Short reports- Technical specifications - Catalogues reading - Presentation techniques. **Ref.** 1. Gerson, S.J., and Gerson, S.M., Technical Communications: Process and product, Pearson/Prentice Hall. 9th Ed., 2017. 2. Gerson, S.J., and Gerson, S.M., Technical writing: Process and product, Pearson/Prentice Hall. 5th Ed., 2006.									
GEN1009	**Problem Solving & Decision-making Skills**	None	2	2				2		مهارات حل المشكلات واتخاذ القرار
	Course Content: Characteristics, environment, and skills necessary for effective problem solving. Problem solving heuristics: Problem definition, Generating solutions, Deciding the course of actions, Implementing the solution, Evaluating the solution. Real-life problem solving. Team work tools and skills. Engineering ethics and responsibility. **Ref.** 1. Fogler, H.S. - LeBlanc, S. E.- Rizzo B. R., Strategies for Creative Problem Solving, 4th Ed, Prentice Hall, 2017.									
GEN1010	**Introduction to Marketing**	None	2	2				2		مقدمة فى التسويق
	Course Content: Introduction – organization of marketing – customer behavior study - market study – products strategy – pricing methods – distribution and marketing methods for products – public relations management. **Ref.** 1. Jim Sterne, Artificial Intelligence for Marketing, Practical Applications, Wiley and SAS Business Series, 2017.									

GEN1011	**Introduction to Economics**	None	2	2				2		**مقدمة في الاقتصاد**
	Course Content: Fundamentals of engineering economy. Time value of money. Evaluation of alternatives. Replacement and retention analysis. Break even analysis. Depreciation methods **Ref.** 1. Leland Blank & Anthony Tarquin, Basics of Engineering Economy, McGRAW-Hill, 2014.									
GEN1012	**Innovation and Entrepreneurship**	None	2	2				2		**الابتكار وريادة الأعمال**
	Course Content: Entrepreneurial thinking; innovation management; opportunity spotting and evaluation; industry and market research; business strategy; business models and business plans; financial forecasting and entrepreneurial finance; pitching to resource providers and negotiating deals; and launching new ventures.									
GEN1013	**Italian Language**	None	2	2				2		**اللغة الايطالية**
	Course Content: Introduction - properties of technical Italian - revision of language grammar - grammar of active sentences and its properties - recognition of some common mistakes in Italian technical sentences - paragraphs - Methods of discussing the main idea - types of paragraphs - reading and analysis of some samples in technical writing different engineering branches to develop the communication skills - translation.									

المحتوى العلمى للمقررات الاختيارية لمتطلبات المجال

Contents of Elective Courses for Discipline Requirements

Course Code	Course Name	Prerequisite	Cr. Hours	Contact Hours					ECTS	المقرر الدراسي
				Lec.	*Studio/ tutorial*	*Lab*	*Work shop*	**Total**		
ECE1609	**Industrial Electronics**	**ECE1603**	**3**	**2**	**2**			**4**		**إلكترونيات صناعية**
	Course Content: Solid-state devices and circuits, Programmable controllers, Thyristors, Lasers, Fiber optics, Power supplies, Op-amp circuits, Open- and closed-loop feedback systems, Input devices, Output devices, AC and DC motors, Motor control devices, Robots and other motion control systems, Data communications. **Ref.** **1.** Bogdan M. Wilamowski, J. David Irwin, "Fundamentals of Industrial Electronics", CRC Press, 2nd Ed., 2017. **2.** Biswanath Paul, "Industrial Electronics and Control", PHI Learning, 3rd Ed., 2014.									
POW1610	**Power Electronics**	**POW1602**	**3**	**2**	**2**			**4**		**إلكترونيات القوى**
	Course Content: Semi-conductor devices as switching elements-static and dynamic characteristics of the thyristor-Thyristors in circuits-cooling. Single-phase controlled rectifier: Analysis of single-phase half wave and full wave circuits including semi and fully controlled bridges - Rectifier with center-tapped transformer. Three-phase controlled rectifiers: Analysis of three-phase half wave and full wave fully controlled circuit.AC voltage regulators: Analysis of single-phase circuits with resistive and inductive loads. DC choppers: Performance and analysis of different classes of DC chopper circuits**.** Inverters: Analysis of single-phase half bridge and full bridge circuits- Three-phase inverter circuits. **Ref.** **1.** Weidong Xiao, "Power Electronics Step-by-Step: Design, Modeling, Simulation, and Control", McGraw Hill, 2021 2. Robert W. Erickson, Dragan Maksimovic', "Fundamentals of Power Electronics", SPRINGER, 3rd Ed., 2020									
ECE1611	**Laser and its Applications**	**ECE1603**	**3**	**2**	**2**			**4**		**الليزر وتطبيقاته**
	Course Content: Introduction: how light is generated, outline and need for the laser, Interaction of EM Radiation with Matter: two-level system, spectral line-shapes, finite lifetime, Doppler effects, absorption and decay processes, spontaneous and stimulated emission, Amplification Criteria: amplification conditions, Lorentzian line-shapes, Gaussian line-shapes, simple cavity model, Fabry-Perot cavity: optics of Fabry-Perot cavity, laser use of Fabry-Perot, laser gain conditions, laser modes, homogeneous broadening, inhomogeneous broadening, control of modes, examples of lasers, Four level laser: four level rate equations, four level gain profile, simple homogeneous laser, output behaviour and power, optimal output conditions, inhomogeneous laser, Laser Modes and Mode Locking: properties of a single mode, multi-mode laser, two-mode system, mode locking in multi-mode laser, mode locking of real laser, active mode locking, passive mode locking, the Kerr lens, Gas Lasers: operation and									

	characteristics of the He-Ne laser and argon ion laser. Summary of other gas lasers, Solid State Lasers: laser media and the role of phonons; titanium sapphire laser; neodymium YAG and glass lasers. Q-switching, Cavity Stability: laser stability conditions and practical laser cavities, Gaussian Beams: scalar potentials, plane wave solution in optical cavity, Gaussian solution, divergence angle and beam parameters, beam waist and Rayleigh region, Gaussian beams in cavities, higher order modes and basic optics of Gaussian beams, Applications Review. **Ref.** **1.** Chunlei Guo, Subhash Chandra Singh, "Handbook of Laser Technology and Applications". Four Volume Set, CRC Press, 2021. **2.** Leon Goldman, "Applications of the laser", CRC Press, 2018.									
CSE1612	**Computer Architecture and Organization**	**GEN0006**	**3**	**2**	**2**			**4**		**هيكلة وتنظيم الحاسب**
	Course Content: Basic structure and operation of a digital computer; Instruction Set Architecture (ISA); Basic Input/ Output; Computer Performance; design of a basic central processing which includes: computer arithmetic units, data and instruction paths; design of control unit using Hardwired and Microprogrammed methods; memory systems; virtual memory; caches; pipelined computer; characteristics of secondary storage; I/O interfacing. **Ref.** 1. William Stallings, "Computer Organization And Architecture Designing For Performance Eighth Edition", Pearson , 2009									
ECE1613	**Fiber Optics Communications Technology**	ECE1606	**3**	**2**	**2**			**4**		**تكنولوجيا الاتصالات بالألياف الضوئية**
	Course Content: Introduction to fiber-optic communications. Semiconductor diode lasers, internal modulation, electro-optic modulation, coherent and incoherent detection, optical fibers and their properties, optical amplifiers, communication systems, and optical networks. **Ref.** **1.** Govind P. Agrawal, "Fiber-Optic Communication Systems", Wiley, 5th Ed., 2021 **2.** John M. Senior, "Optical Fiber Communications: Principles and Practice", Pearson, 3rd Ed., 2009									
ECE1614	**Antennas**	**ECE1608**	**3**	**2**	**2**	**-**	**-**	**4**		**هوائيات**
	Course Content: Classification of antennas, antenna types, antenna parameters and the radiation characteristics, radiation from an ideal dipole and line source, half-wavelength dipole, small loop antennas, Yagi – Uda antenna, parabolic and Casse grain antennas. Aperture antenna, helical antenna, rectangular									

	microstrip antenna and antenna arrays. **Ref.** 1. C. balanis, "Antenna theory: Analysis & Design," 4th edition, Wiley-Interscience, 2016. 2. Warren L. Stutzman and Gary A. Thiele, Antenna Theory and Design, Wiley; 3 edition , 2012

المحتوى العلمى للمقررات الاختيارية لمتطلبات التخصص

Contents of Elective courses for specialty Requirements

Course Code	Course Name	Prerequisite	Cr.Hours	Contact Hours					ECTS	المقرر الدراسي
				Lec.	*Studio/ tutorial*	*Lab*	*Work shop*	**Total**		
POW2611	**Systems and Applications of SCADA**	POW3604	**3**	**2**	**2**			**4**		**أنظمة وتطبيقات SCADA**
	Course Content: Interpret process drawings, control narratives, trends and basic ladder logic, Differentiate between the different components of a control system (e.g. PLC, instruments, HMI, SCADA), Explain the purpose of the various components within a control system, Operate user interfaces on site and via remote access, Describe control system maintenance and troubleshooting steps, Identify appropriate troubleshooting steps to use for given operational issues, Recommend appropriate solutions to operational issues, Contribute to the design/upgrading of SCADA and control systems. **Ref.** 1. Jacob Brodsky, Robert Radvanovsky, "Handbook of SCADA-control systems security", CRC Press, 2016. 2. David Bailey, Edwin Wright, "Practical SCADA for Industry", Elsevier. 2003. 3. Stuart A. Boyer, "Scada: Supervisory Control and Data Acquisition", Elsevier, 2004.									
POW2612	**Protection of Electric Power Systems**	POW3604	**3**	**2**	**2**			**4**		**حماية نظم القوى الكهربية**
	Course Content: Introduction – Fuse and Circuit Breakers – Protection Relaying- Current and Voltage Transformers – Protection of Feeders and Transmission Lines – protection of AC Machines (Generator and Motors) – Power Transformer Protection – Busbar Protection – static and Digital relays. **Ref.** **1.** John Ciufo, Aaron Cooperberg, "Power System Protection: Fundamentals and Applications", Wiley-IEEE Press, 2021. **2.** Hector J. Altuve Ferrer, Edmund O. Schweitzer III, "Modern Solutions for Protection, Control and Monitoring of Electric Power Systems", Quality Books, Inc., 2010. 3. Leslie Hewitson, Mark Brown, Ramesh Balakrishnan, "Practical power systems protection", Newnes, 2005.									
POW2613	**Rationalization and management of electric energy**	POW3604	**3**	**2**	**2**			**4**		**ترشيد و إدارة الطاقة الكهربية**
	Course Content: Electrical energy management: Power supply side, Electrical energy management (EEM) in consumer side, Role of load demand side management, and Electricity Tariff. **Ref.**									

	1. Craig B. Smith, Kelly Parmenter, "Energy Management Principles, Applications, Benefits, Savings", 2nd Ed., Elsevier, 2015. **2.** Giovanni Petrecca, "Energy Conversion and Management: Principles and Applications", Springer International Publishing, 2014.									
POW2614	**Renewable Energy Systems**	POW3 604	**3**	**2**	**2**			**4**		**منظومات الطاقة المتجددة**
	Course Content: This course introduces the renewable energy to the engineering students. It includes study the environmental impact of using fossil fuel and the importance of using renewable energy sources, different types of renewable energy include, wind energy resources, wind energy turbine construction and operation, types of generators used with wind turbine. Active and passive thermal solar energy systems – Photovoltaic solar cell construction, theory of operation, equivalent circuit, and PV systems – Ocean energy systems, tidal and wave energy – Geothermal energy systems – Different control techniques used with renewable energy systems. **Ref.** **1.** Mamdouh Assad, Marc A. Rosen, "Design and Performance Optimization of Renewable Energy Systems", Academic Press, 2021. **2.** Muhammad Kamran, Muhammad Rayyan Fazal, "Renewable energy conversion systems: Technologies, Design and Operation", Academic Press, 2021. **3.** Mukhtar Ahmad, "Operation and control of renewable energy systems", John Wiley & Sons, 2018.									
POW2615	**Analysis of Electrical Power Systems**	POW3 604	**3**	**2**	**2**			**4**		**تحليل نظم القوى الكهربية**
	Course Content: Complex power flow- Network matrix operation- Methods of Power flow solutions (Gauss-Seidel, Newton Raphson, Fast decouple) - DC load flow- Economic operation of power plants (Thermal and Hydro.)- Fuel rate and incremental fuel cost main constraints- Transmission line losses consideration- Load scheduling – Power system stability- Concept of stability- Transient stability- Swing equation – Equal area criterion- Clearing angle and time- multi-machine system. **Ref.** 1. Abdus Salam, "Fundamentals of Electrical Power Systems Analysis", Springer, 2020 2. P. Venkatesh; B. V. Manikandan; S. Charles Raja; A. Srinivasan, "Electrical Power Systems: Analysis, Security and Deregulation", PHI Learning, 2012.									
POW2616	**Energy Distribution Systems**	POW3 604	**3**	**2**	**2**			**4**		**أنظمة توزيع الطاقة**
	Course Content: Load modeling of distribution networks, Application of distribution transformers, Design of sub-transmission lines, Voltage drop, Power losses, Distribution system voltage regulation, Distribution system protection, Distribution system reliability.									

	Ref. 1. Radian Belu, "Building Electrical Systems and Distribution Networks-An Introduction", CRC Press, 2020. 1. Abdelhay A. Sallam, Om P. Malik, "Electric Distribution Systems", Wiley-IEEE, 2019.									
ECE2617	**Smart Home Systems**	**ECE 2607**	**3**	**2**	**2**			**4**		**أنظمة المنازل الذكية**
	Course Content: Classic smart home overview, Measuring home conditions, managing home appliances, Controlling home access, The main components of smart home. Cloud computing and its contribution to IoT and smart home, Centralized event processing, a rule-based system, Event processing languages. Rediscovering workflow from events, Practical aspects and implementation considerations for IoT and smart home, Smart home and IoT examples: Discovery of water leaks and its prevention, Smoke detectors, Incident management to control home appliances. **Ref.** 1. Mahmoud A. Al-Qutayri , "Smart Home Systems", In-The, 2010 2. Zeeshan Ahmed, Mujtaba Ali, "Smart House: Artificially Intelligent Home Automation System", VDM, 2011.									
ECE2618	**Audio Signals Processing**	**ECE2 601**	**3**	**2**	**2**			**4**		**معالجة إشارات الصوت**
	Course Content: Sampling and quantization, analog to digital converters, time and frequency domains, spectral analysis, vocoding, digital filters, audio effects, music audio analysis and synthesis, and other advanced topics in audio signal processing. Implementation of algorithms using Matlab and dedicated DSP platforms is emphasized. **Ref.** **1.** Trevor J. Cox, Francis F Li, "Digital signal processing in audio and acoustical engineering", CRC Press, 2019. 2. Udo Zölzer, "Digital Audio Signal Processing" Wiley, 2008.									
CSE2619	**Software Engineering**	**None**	**3**	**2**	**2**			**4**		**هندسة البرمجيات**
	Course Content: Introduction to software engineering, Importance of software. Software development life-cycle: Requirement analysis and engineering, Architecture and software design, coding, testing and maintenance etc. Software Processes: Introduction, Software Process Models (Waterfall Model, and others), Software Design and Implementation. Functional & Non-functional Requirements. UML diagrams. Verification and Validation software and System test. Learn how to manage projects (PM). **Ref.** 1. Bhuvan Unhelkar , " Software Engineering with UML ", Auerbach Publications , 2017									

	Web Design	**CSE 1604**	**3**	**2**	**2**			**4**		**تصميم المواقع الالكتروني ة على الانترنت**
CSE2620	**Course Content:** 1. Overview and Evolution of Web Engineering • Simple and Static Hypermedia Documents • multi-domain platform for Dynamic Web Applications • Enterprise Driven Web-Enabled Applications 2. Web Technologies • HTTP: Enabling Content (HTML) and Design (CSS) • Dynamic HTML and Client-Side Business Logic • Dynamic Web Pages and Server-Side Business Logic • Overview of eXtensible Markup Language (XML) • Overview of Web Services and Remote Business Logic 3. The Development Process • Decomposing the Software Development Process • Structuring the Software Development Process • Web-Specific Software Development Processes • Examples of Web- Specific Development Processes 4. Requirements Engineering • Web Requirements Engineering Concepts • Organization Requirements Analysis • Application Domain Analysis • Navigation and Interaction Analysis 5. Web Application Design • Design Concepts: Principles and Process • Design Types and Models: Workflow Design, Data Design, Navigation Design, Presentation Design, Architecture Design • Model-Driven Engineering and Web Engineering 6. Implementation, Deployment, and Maintenance • Implementing the Presentation Layer • Deployment and Installation • Maintenance and Evolution: The Role of Model-Driven Design and Industry Solutions 7. Quality Assessment • Quality Models • Testing and Usability Evaluation • Web Design Methods and Quality Assessment • Automatic Tools for Testing and Usability evaluation. **Ref.** 1. Terry Felke-Morris, “Basics of Web Design: HTML5 & CSS”, 5th Ed., Pearson, 2019. 2. Klaus-Dieter Schewe, Bernhard Thalheim, “Design and Development of Web Information Systems”, Springer Berlin Heidelberg, 2019.									
	Digital Image Processing	**ECE2 601**	**3**	**2**	**2**			**4**		**معالجة الصور الرقمية**
CSE2621	**Course Content:** Domains of image processing: (the spatial frequency domain, the sampling theorem, template matching and the convolution theorem, spatial filtering). Image Enhancement: (Pixel Transformations: - basic transformations, - histogram stretching, - histogram equalization; filters, linear filtering in the space domain, linear filtering in the frequency domain). Image Restoration: (noise and noise models, noise removal by linear and nonlinear filtering techniques in spatial and frequency domain, degradation models, Inverse filtering, Wiener filtering). Image segmentation. Image Compression: (Compression model, different coding techniques and compression algorithms). **Ref.** **1.** Rafael Gonzalez, Richard Woods, Stevens Eddins - Digital Image Processing Using MATLAB-Gatesmark Publishing (2020) **2.** Rafael C. Gonzalez, Richard E Woods, “Digital image processing”, Pearson, 2018.									
CSE2622	**Mobile**	**ECE**	**3**	**2**	**2**			**4**		**برمجة**

	Devices Programming	**2604**								**الاجهزة المحمولة**
	Course Content: Introduction, Survey existing mobile applications, Mobile operating systems, Programming Environments, Development languages, Installation, Coding, Native code, Emulators, Other programming options, Mobile Device Features, Mobile devices vs. PCs. Mobile device features not available on PCs, Accelerometer, GPS, User Interface Design. **Ref.** 1. Tommi Mikkonen, "Programming Mobile Devices: An Introduction for Practitioners", Wiley, 2007. 2. Reto Meier, "Professional Android 4 Application Development", Wiley India, 2012. 3. James C Sheusi, "Android Application Development for Java Programmers", Cengage Learning, 2013.									
CSE2623	**Data Mining**	CSE 2609	**3**	**2**	**2**			**4**		**تنقيب في البيانات**
	Course Content: Data Mining studies algorithms and computational paradigms that allow computers to find patterns and regularities in databases, perform prediction and forecasting, and generally improve their performance through interaction with data. It is currently regarded as the key element of a more general process called Knowledge Discovery that deals with extracting useful knowledge from raw data. The knowledge discovery process includes data selection, cleaning, coding, using different statistical and machine learning techniques, and visualization of the generated structures. **Ref.** 1. Ian H. Witten and Eibe Frank, Data Mining: Practical Machine Learning Tools and Techniques (Second Edition), Morgan Kaufmann, 2005, ISBN: 0-12-088407-0.									
CSE2624	**Machine Learning**	CSE 2610	**3**	**2**	**2**			**4**		**تعليم الالة**
	Course Content: This course provides exploration of machine learning including how to design, implement, and evaluate machine learning algorithms to solve real-world problems. Also, it covers a range of topics including neural networks, deep learning, reinforcement learning, Feedforward Neural Networks, Regularization and Optimization, Convolutional Neural Networks, Recurrent Neural Networks, Generative models, deep reinforcement learning, and other advanced neural network techniques. **Ref.** 1. Charu C. Aggarwal, " Neural Networks and Deep Learning: A Textbook", Springer, 2018. 2. Ian Goodfellow, Yoshua Bengio, and Aaron Courville," Deep Learning", MIT Press , 2016. 3. Andreas C. Müller and Sarah Guido, " Introduction to Machine Learning with Python: A Guide for Data Scientists", 2016.									
CSE2625	**Parallel and Distributed Systems**	**CSE 2604**	**3**	**2**	**2**			**4**		**الأنظمة المتوازية والموزعة**
	Course Content:									

	Introduction to Parallel computers models. Performance and scalability of parallel systems. Dense Matrix Algorithm. Searching. Sorting, Graph Algorithm. MPI programming, Cluster, Grid, and Cloud computing **Ref.** 1. Yong Zhang, Yicheng Xu, Hui Tian, "Parallel and Distributed Computing, Applications and Technologies ", 2021.									
ECE3612	**Wireless Sensors Networks**	**ECE3 601**	**3**	**2**	**2**			**4**		**شبكات المجسات اللاسلكية**
	Course Content: Introduction to wireless networks, architectures and technologies, Wireless sensor network platforms: Hardware and Software, Communication architecture and protocols for WSN (MAC, Link, Routing), Energy management, Sensor data acquisition, processing and handling, Signal processing, target localization and tracking, self-organization, Modeling and Simulation of WSN. **Ref.** **1.** Ibrahiem M. M. El Emary, S. Ramakrishnan, "Wireless Sensor Networks From Theory to Applications", CRC Press, 2016. **2.** Ian F. Akyildiz, Mehmet Can Vuran, "Wireless Sensor Networks", Wiley, 2010. **3.** Anna Hac, "Wireless Sensor Network Designs", Wiley, 2003.									
ECE3613	Smart Cities	**ECE3 607**	**3**	**2**	**2**			**4**		**المدن الذكية**
	Course Content: The characteristics of a smart city, The drivers of smart cities, How to identify the essential technologies in a smart city, The impacts of Internet of Things (IoT) on cities, The dimensions of smart government in a smart city, Main concerns and challenges in smart city developments, Review case studies of smart city projects around the world. **Ref.** 1. Juan Carlos Augusto, "Handbook of Smart Cities", Springer, 2021). 2. Germaine Halegoua, "Smart Cities", The MIT Press, 2020. 3. Fadi Al-Turjman, "Intelligence in IoT-enabled smart cities", Taylor & Francis CRC, 2019.									